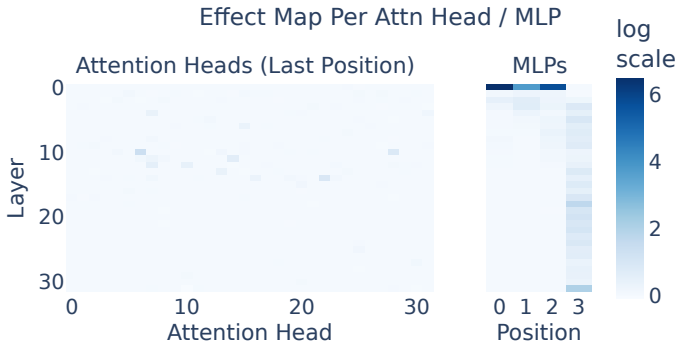
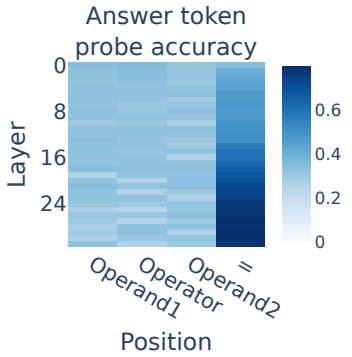
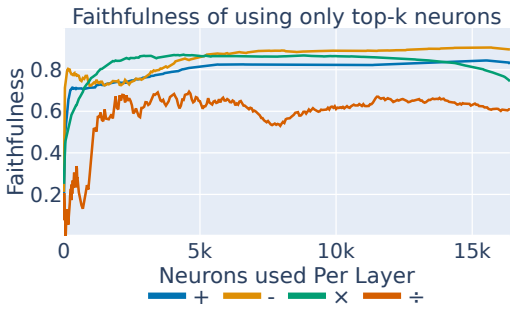


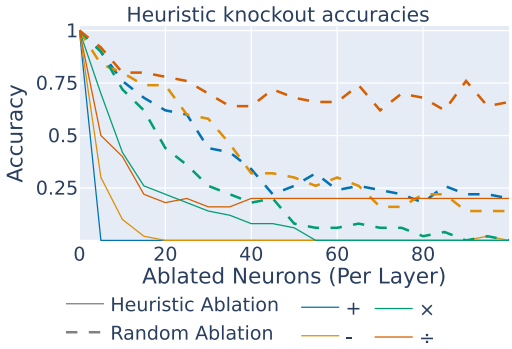
(a) Pythia-6.9B localization results.



(b) Pythia-6.9B linear probing results.

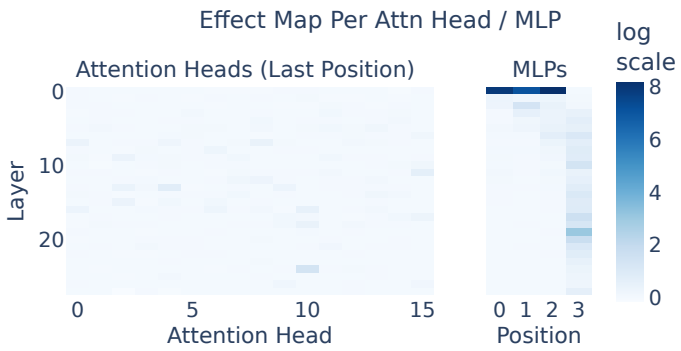


(c) Pythia-6.9B faithfulness as function of MLP neurons in each layer.

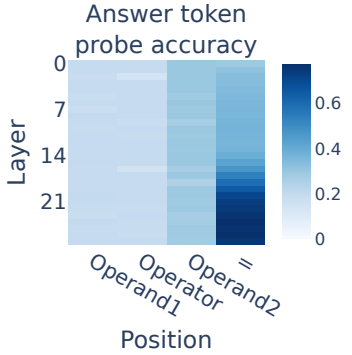


(d) Pythia-6.9B prompt-specific heuristic knockout.

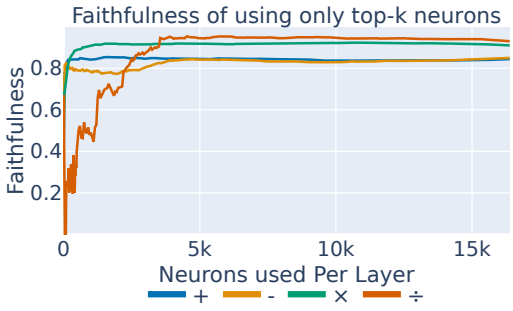
Figure 17: Results for all analyses on Pythia-6.9B.



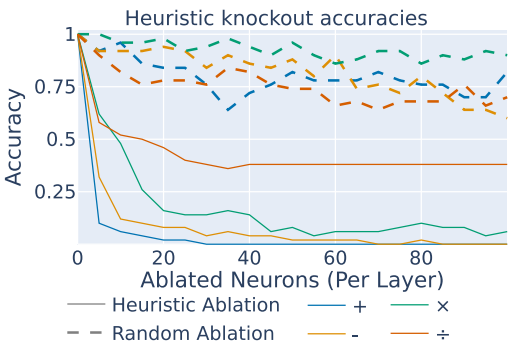
(a) GPT-J localization results.



(b) GPT-J linear probing results.



(c) GPT-J faithfulness as function of MLP neurons in each layer.



(d) GPT-J prompt-specific heuristic knockout.

Figure 18: Results for all analyses on GPT-J.

Table 4: Examples of heuristic neurons for the addition operator

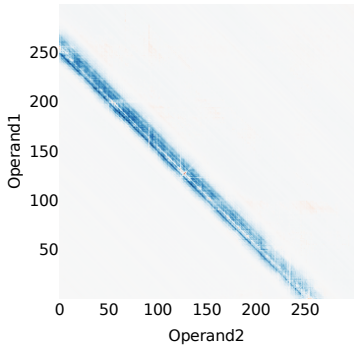
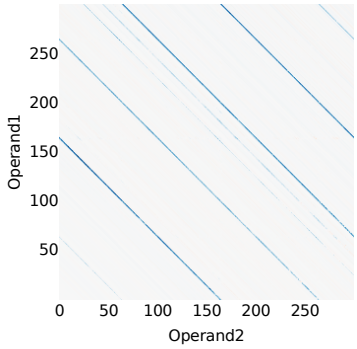
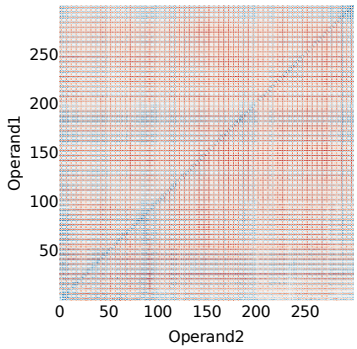
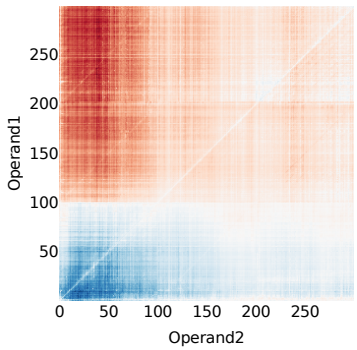
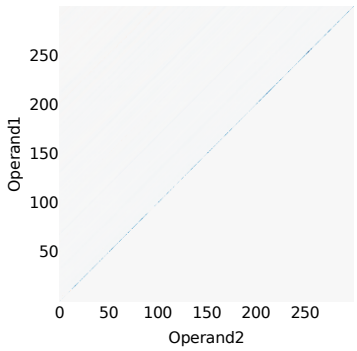
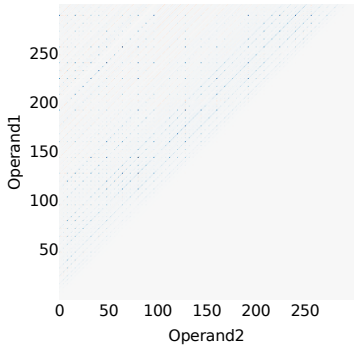
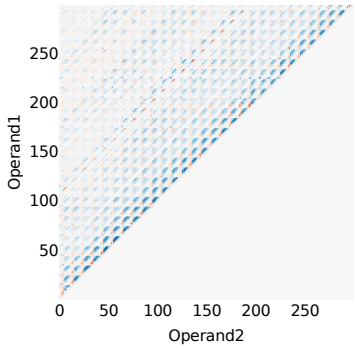
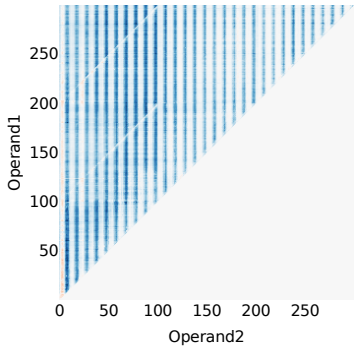
Neuron number $\mathbf{h}_{post}^{l,n}$	Activation pattern	Top-10 numerical logits in value vector $\mathbf{v}_{out}^{l,n}$	Classified Heuristics
$\mathbf{h}_{post}^{31,9208}$		'254', '255', '253', '256', '250', '257', '252', '251', '258', '259'	$\text{range}_{result} \in [240, 270]$, $\text{range}_{result} \in [250, 300]$, pattern_{result} .5.
$\mathbf{h}_{post}^{31,2919}$		'164', '163', '464', '363', '63', '063', '364', '963', '263', '663'	pattern_{result} .6., pattern_{result} .64
$\mathbf{h}_{post}^{17,1463}$		'2', '542', '792', '979', '372', '502', '882', '962', '602', '02'	$op1 \equiv 2 \pmod 5$, $op2 \equiv 2 \pmod 5$
$\mathbf{h}_{post}^{16,10226}$		'922', '169', '322', '258', '873', '753', '129', '852', '41', '33'	$\text{range}_{op1} \in [0, 50]$, $\text{range}_{op1} \in [0, 100]$, $\text{range}_{op2} \in [0, 100]$

Table 5: Examples of heuristic neurons for the subtraction operator

Neuron number $\mathbf{h}_{post}^{l,n}$	Activation pattern	Top-10 numerical logits in value vector $\mathbf{v}_{out}^{l,n}$	Classified Heuristics
$\mathbf{h}_{post}^{30,8806}$		'0', '000', '00', '006', '002', '004', '001', '005', '003', '009'	identical operands, pattern_{result} 000
$\mathbf{h}_{post}^{29,1647}$		'192', '384', '288', '576', '336', '672', '96', '224', '608', '768'	$result \equiv 0 \pmod 8$
$\mathbf{h}_{post}^{16,747}$		'15', '5', '015', '535', '764', '3', '762', '021', '4', '13'	pattern_{result} 0.5
$\mathbf{h}_{post}^{16,360}$		'369', '271', '761', '762', '522', '918', '972', '783', '325', '531'	pattern_{op2} ..8